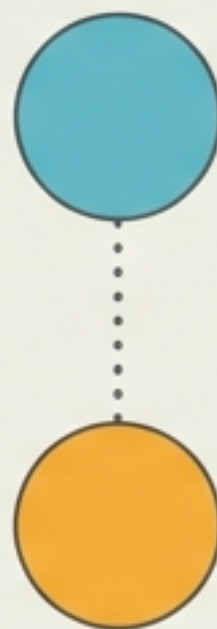


SYSTEM STATUS

TARGET: LATENT VARIABLES
PROTOCOL: PROBABILISTIC INFERENCE
STATUS: READY



UNVEILING THE HIDDEN

A DATA DETECTIVE'S GUIDE TO
HIDDEN MARKOV MODELS (HMMs)

The Core Intuition: What We See vs. What Is Real

Observable Sequence (What we measure)



WORDS / SENTENCES

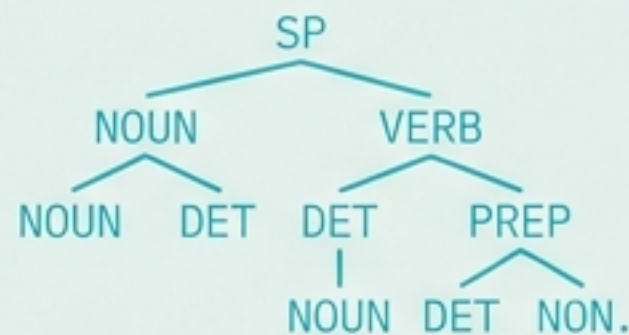


MARKET TICKER



UMBRELLAS / RAIN

Hidden Sequence
(The underlying truth)



SYNTAX TREES

Emissions

Emissions

Emissions



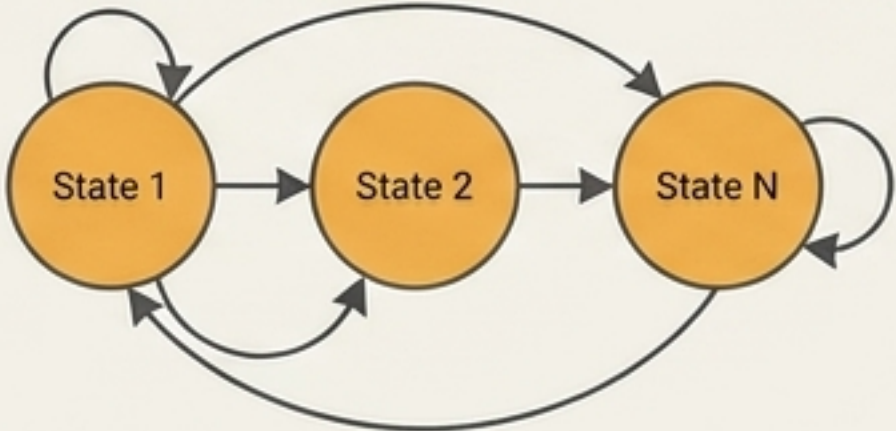
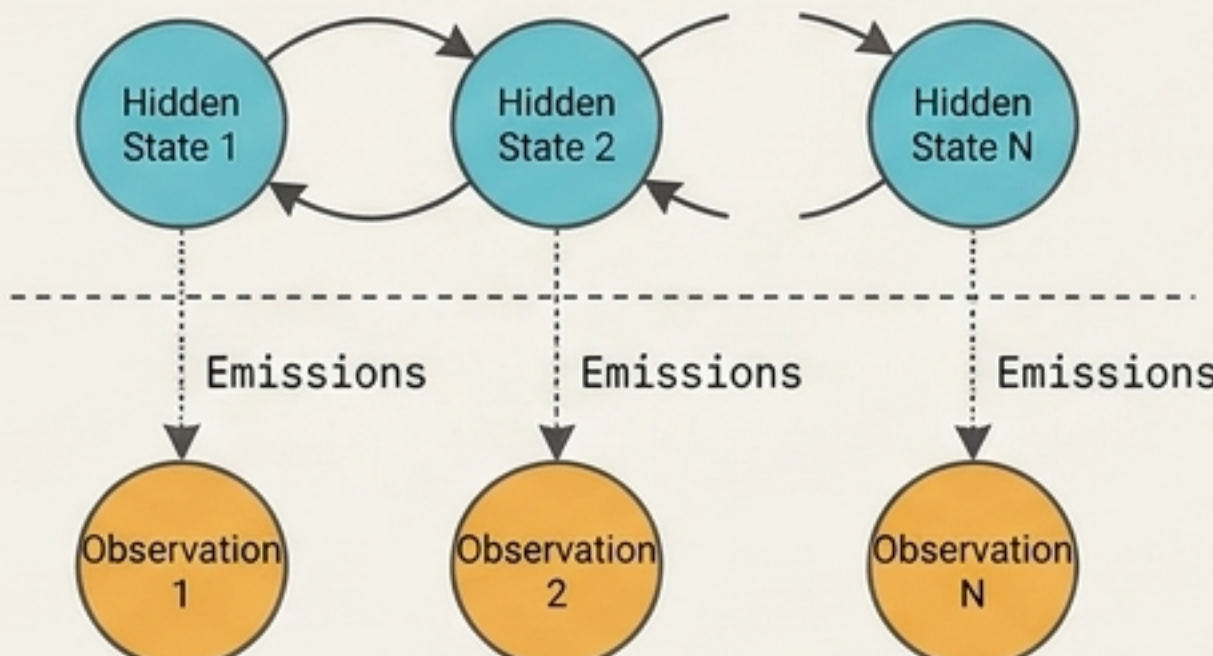
MARKET MOOD (BULL/BEAR)



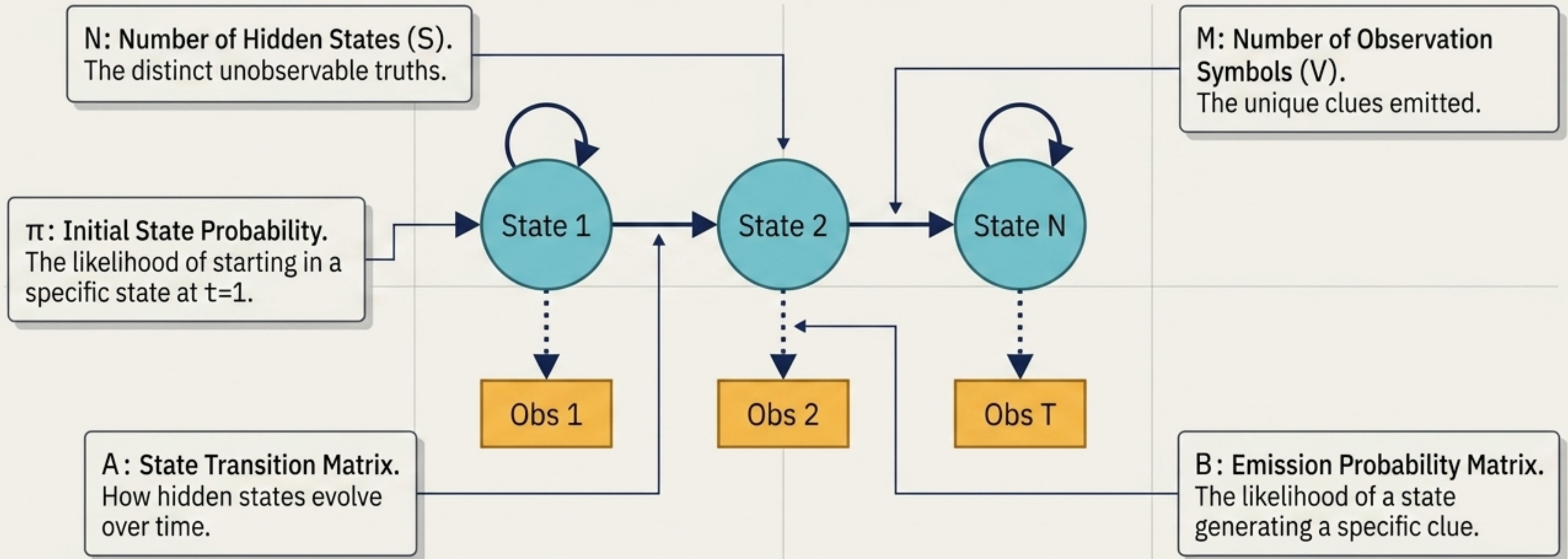
WEATHER SYSTEMS (CLOUDS/SUN)

An HMM acts as a statistical detective, modeling the relationship between the unseen truth and the observable clues to reverse-engineer reality.

The Foundational Leap: Markov Chains vs. HMMs

	Markov Chain	Hidden Markov Model
Node Structure		
State Visibility	Visible to the observer.	Completely obscured.
Source of Observation	The state itself is the observation.	The state emits a separate observation.
Weather Analogy	Directly recording 'Sunny' or 'Rainy'.	Recording 'Umbrella' or 'Sunglasses' to guess the weather.

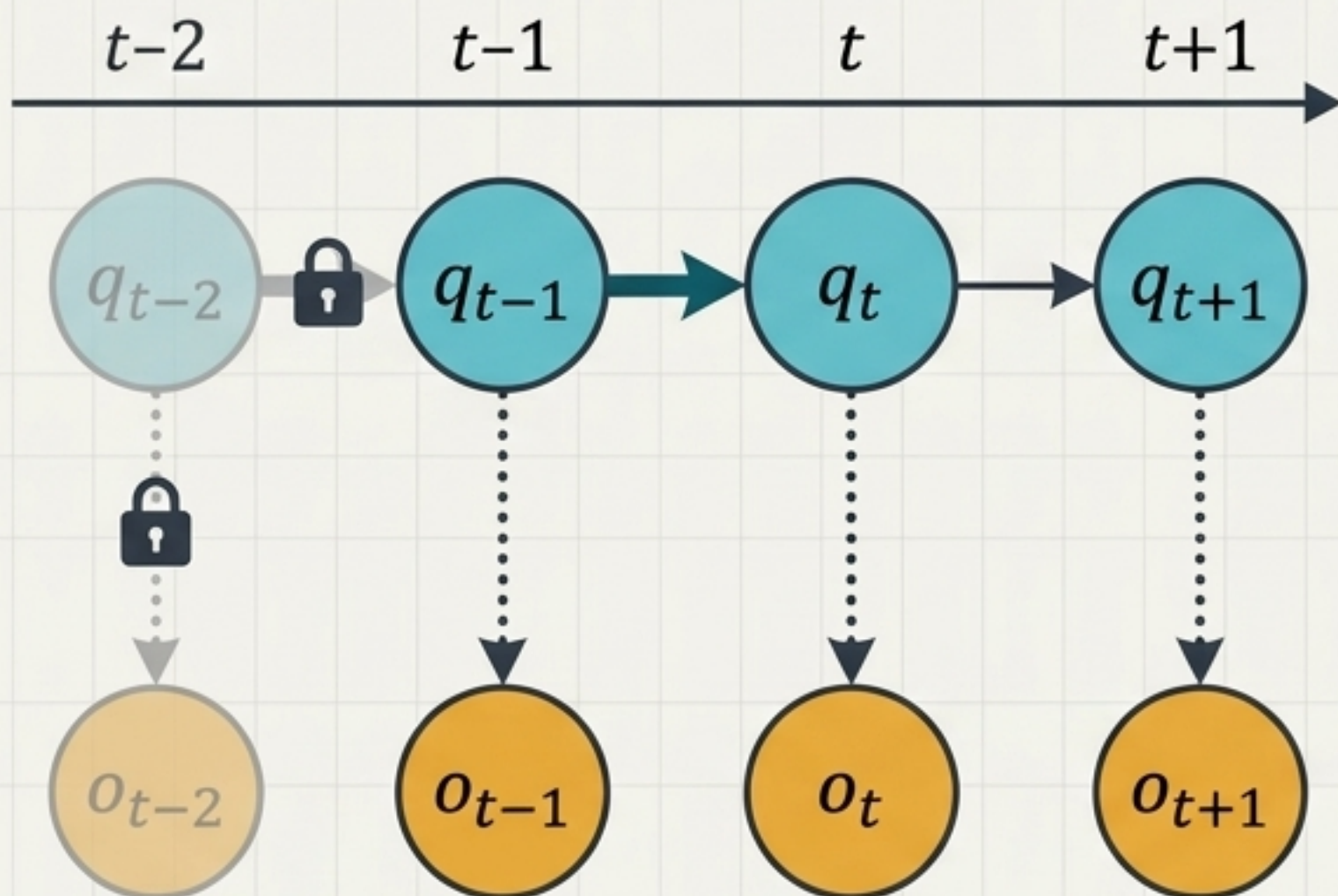
Anatomy of an HMM: The System Parameters



The full system is defined as $\lambda = (A, B, \pi)$

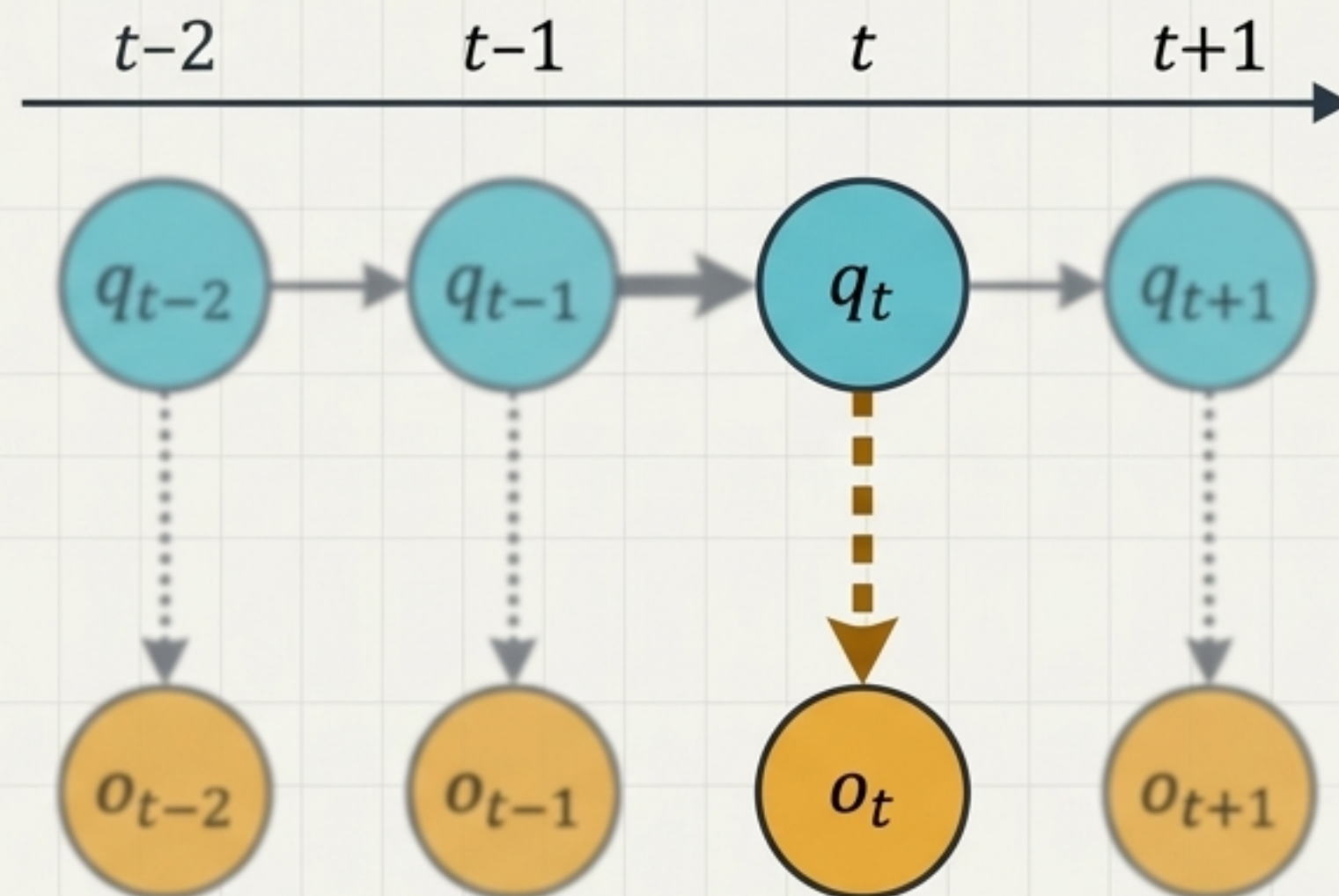
The Boundary Conditions: Two Foundational Assumptions

Markov Assumption (Limited Horizon)



The next state depends only on the current state.
The deep past is forgotten.
 $P(q_t|q_{t-1})$.

Output Independence Assumption



An observed clue depends only on its immediate hidden state. It is blind to the rest of the timeline.
 $P(o_t|q_t)$.

The Diagnostic Matrix: Three Fundamental Problems

Evaluation (Likelihood)	Decoding	Learning (Training)
Input: Model (λ) + Observation (O) The Question: What is the probability $P(O \lambda)$ that this exact model generated these exact clues? The Algorithm: The Forward Algorithm	Input: Model (λ) + Observation (O) The Question: What is the single most likely hidden sequence (Q) that produced these clues? The Algorithm: The Viterbi Algorithm	Input: Observation Sequence (O) only The Question: How do we adjust the parameters (λ) to maximize probability? The Algorithm: The Forward-Backward (Baum-Welch) Algorithm

Case File 01: Jason's Ice Cream & The Hidden Weather

Narrative Briefing

You are an observer.
You cannot see the weather outside. You can only see how many ice creams Jason eats each day.

Can we calculate the likelihood of a specific eating streak based on weather patterns?

System Initialization

Hidden States (N=2):

Hot (H)

Cold (C)

Observations (M=3):

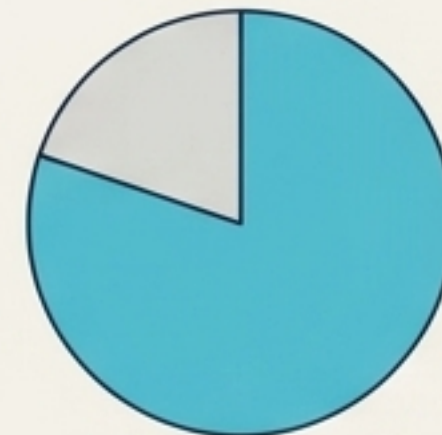
1 Ice Cream

2 Ice Creams

3 Ice Creams

Initial Probabilities (π):

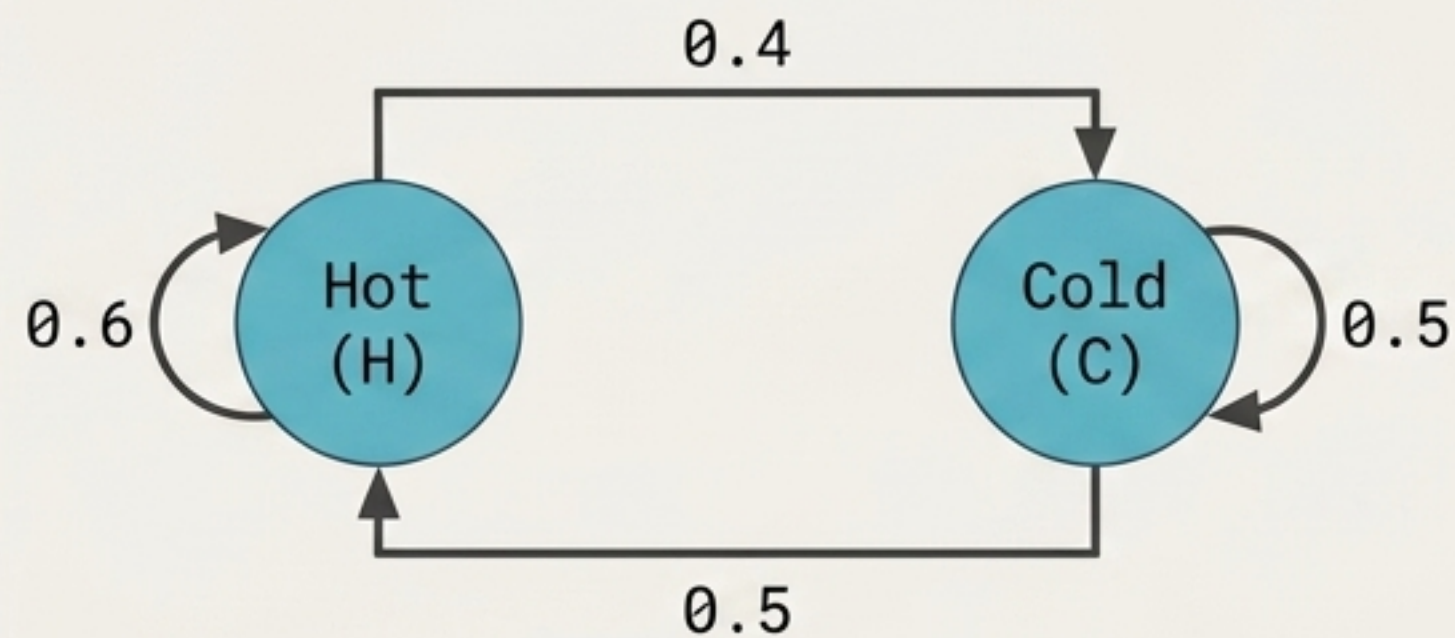
$P(\text{Cold}) = 0.2$



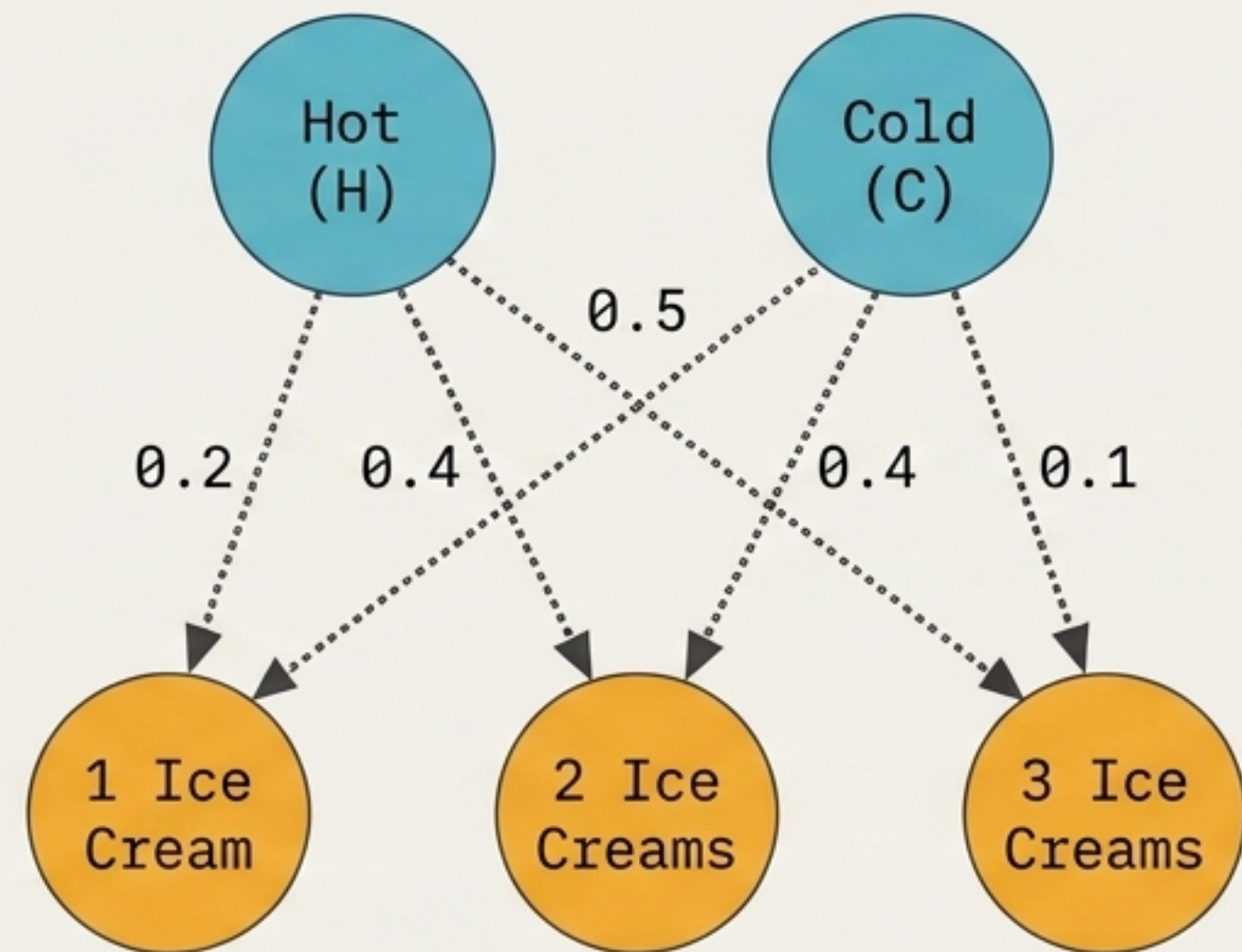
$P(\text{Hot}) = 0.8$

System Calibration: Transition & Emission Probabilities

A Matrix: Weather Transitions



B Matrix: Ice Cream Emissions



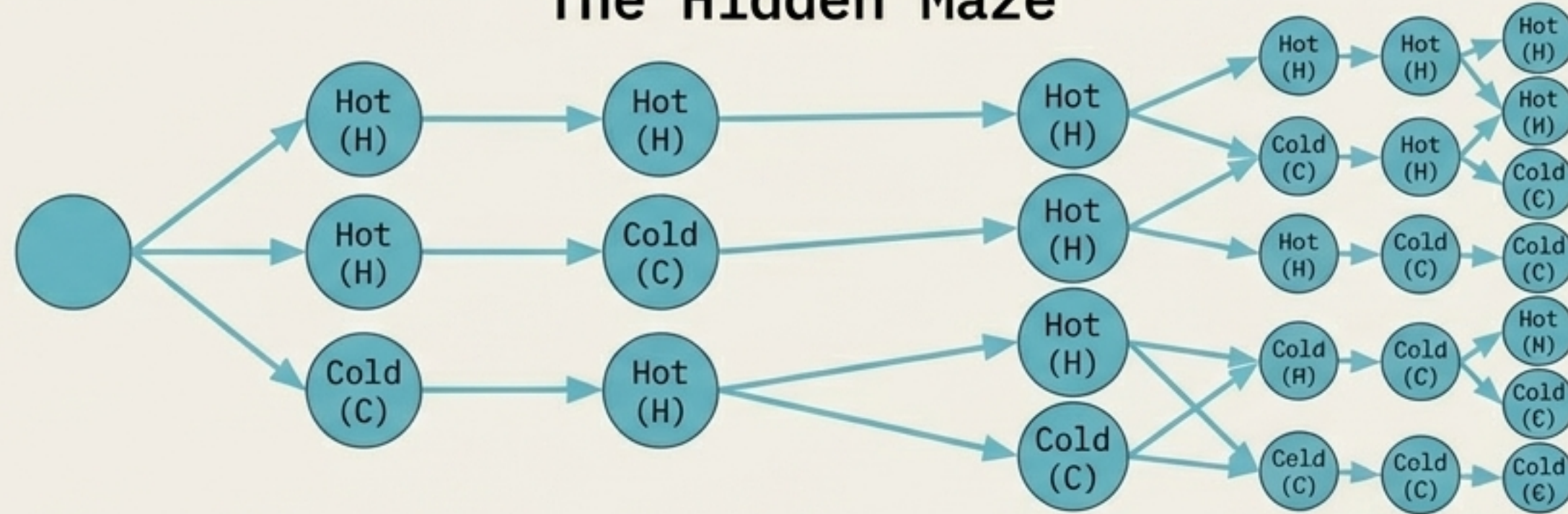
Notice all probability rows sum to 1.0. The system is perfectly constrained.

The Combinatorial Bottleneck

Target Sequence: $O = (3, 1, 3)$

Jason eats 3, then 1, then 3 ice creams.

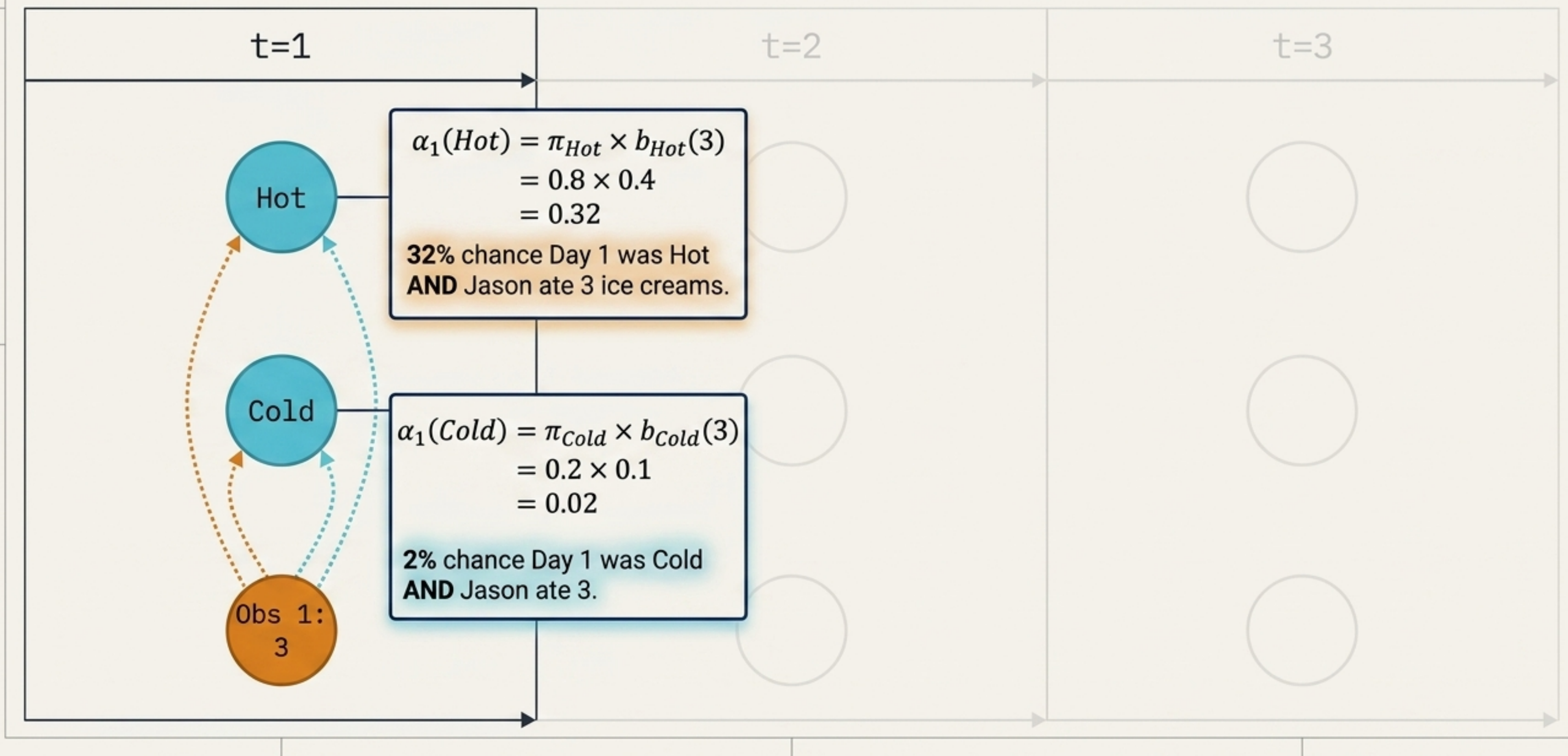
The Hidden Maze



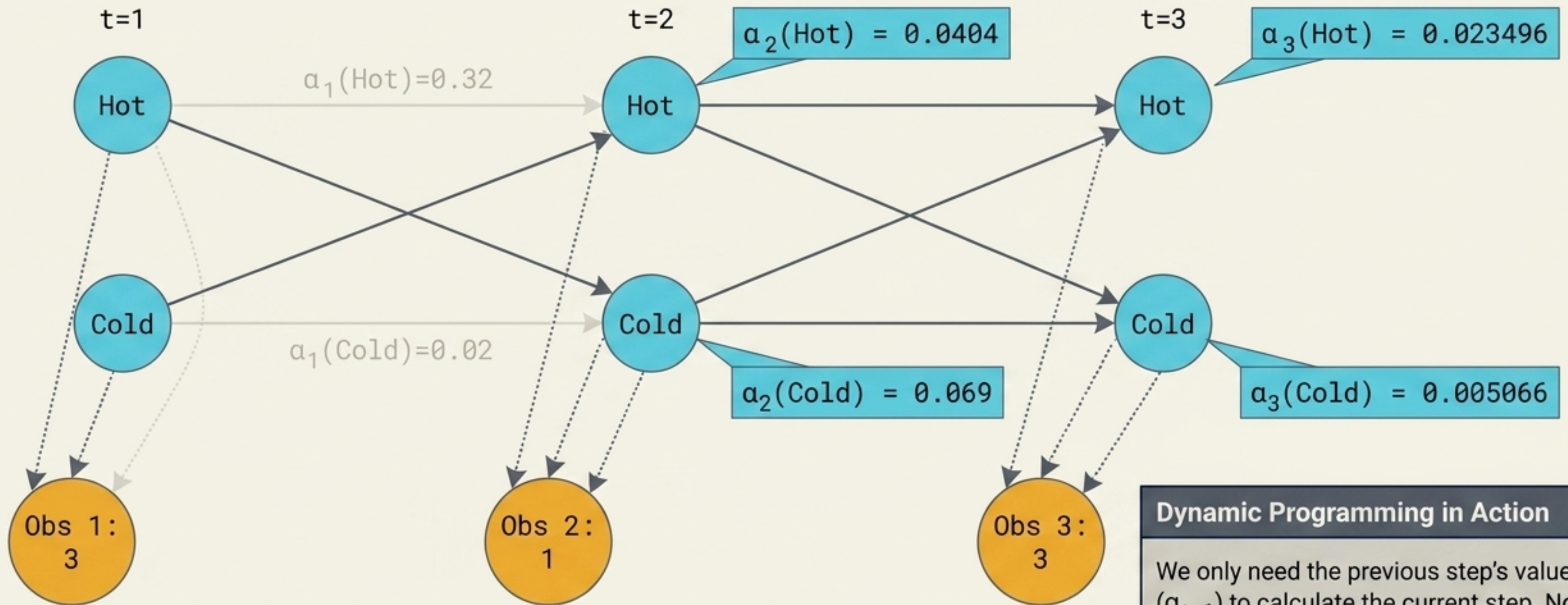
Calculating $P(O|\lambda)$ manually requires checking N^T possible hidden paths.
For long sequences, this is computationally impossible.

The Solution: Enter the Forward Algorithm. A dynamic programming approach.

The Forward Algorithm | Step 1: Initialization (t=1)



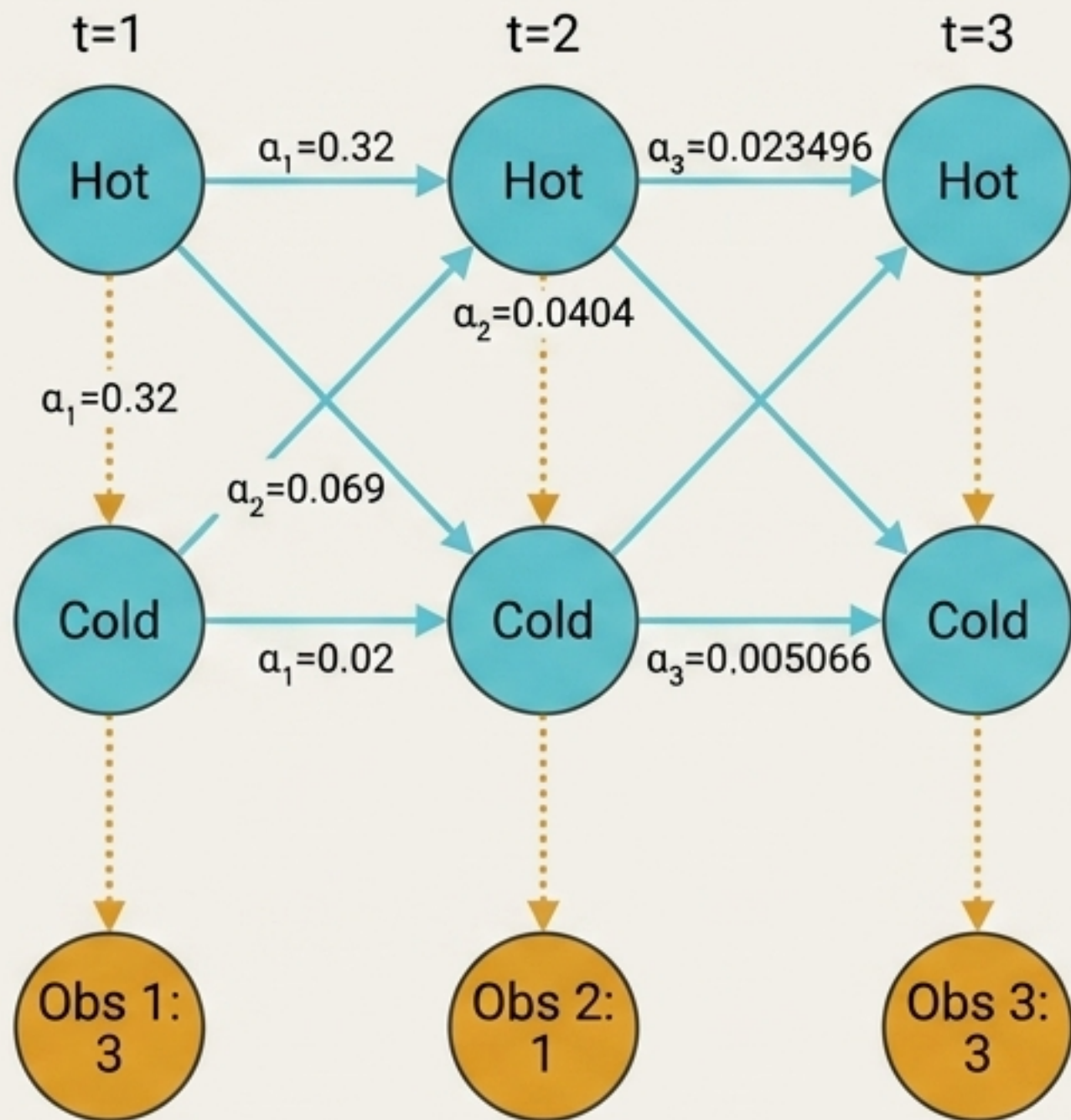
The Forward Algorithm | Step 2: Recursion (t=2 to t=3)



Dynamic Programming in Action

We only need the previous step's values (α_{t-1}) to calculate the current step. No redundant calculations.

The Forward Algorithm | Step 3: Termination



Final Calculation

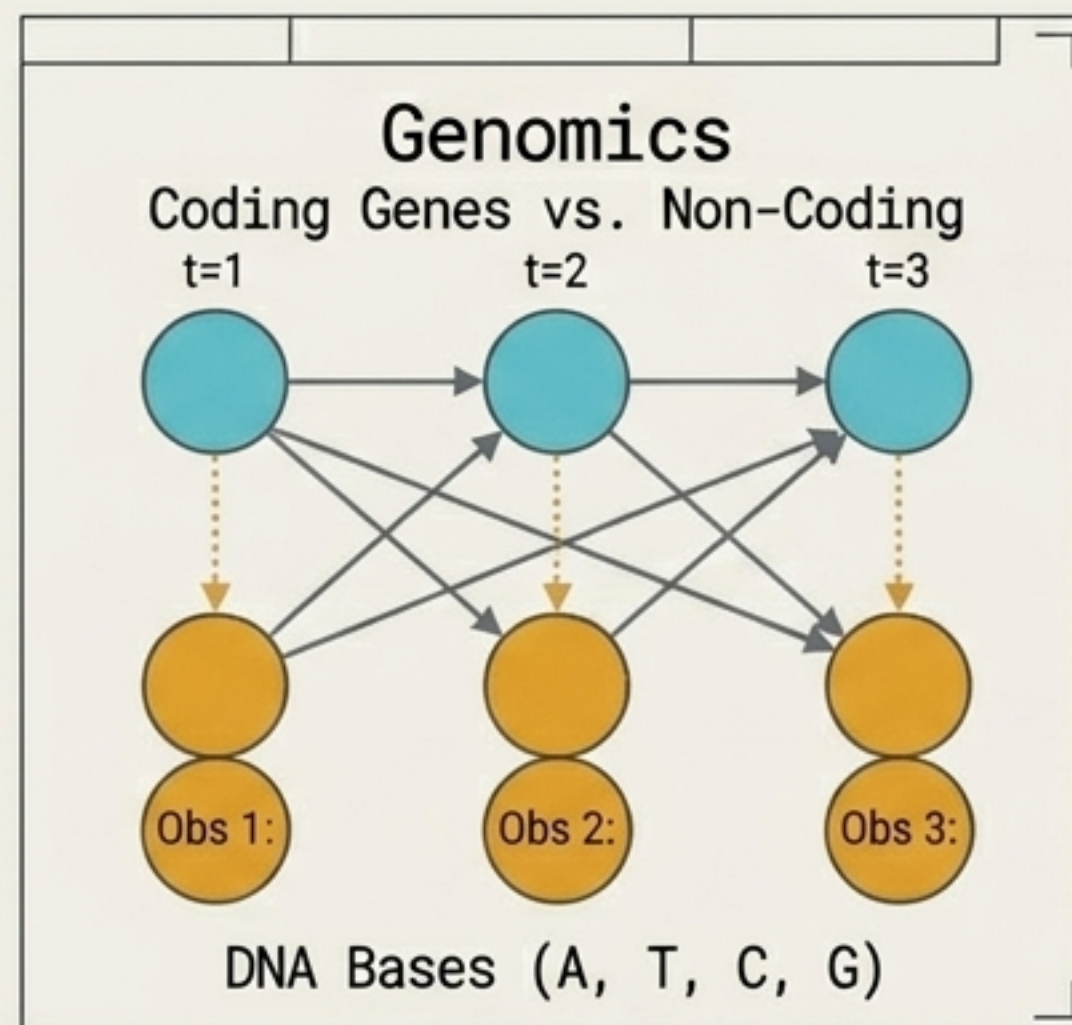
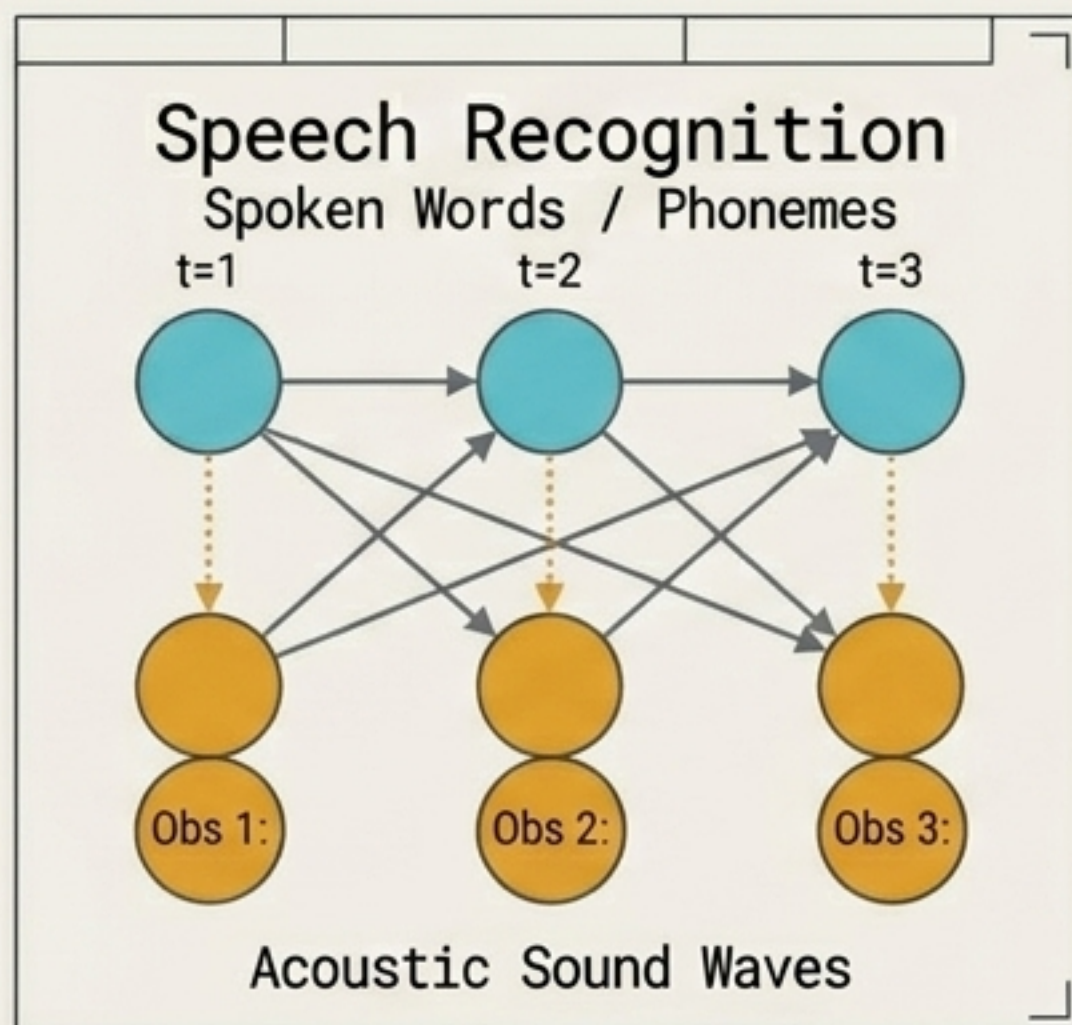
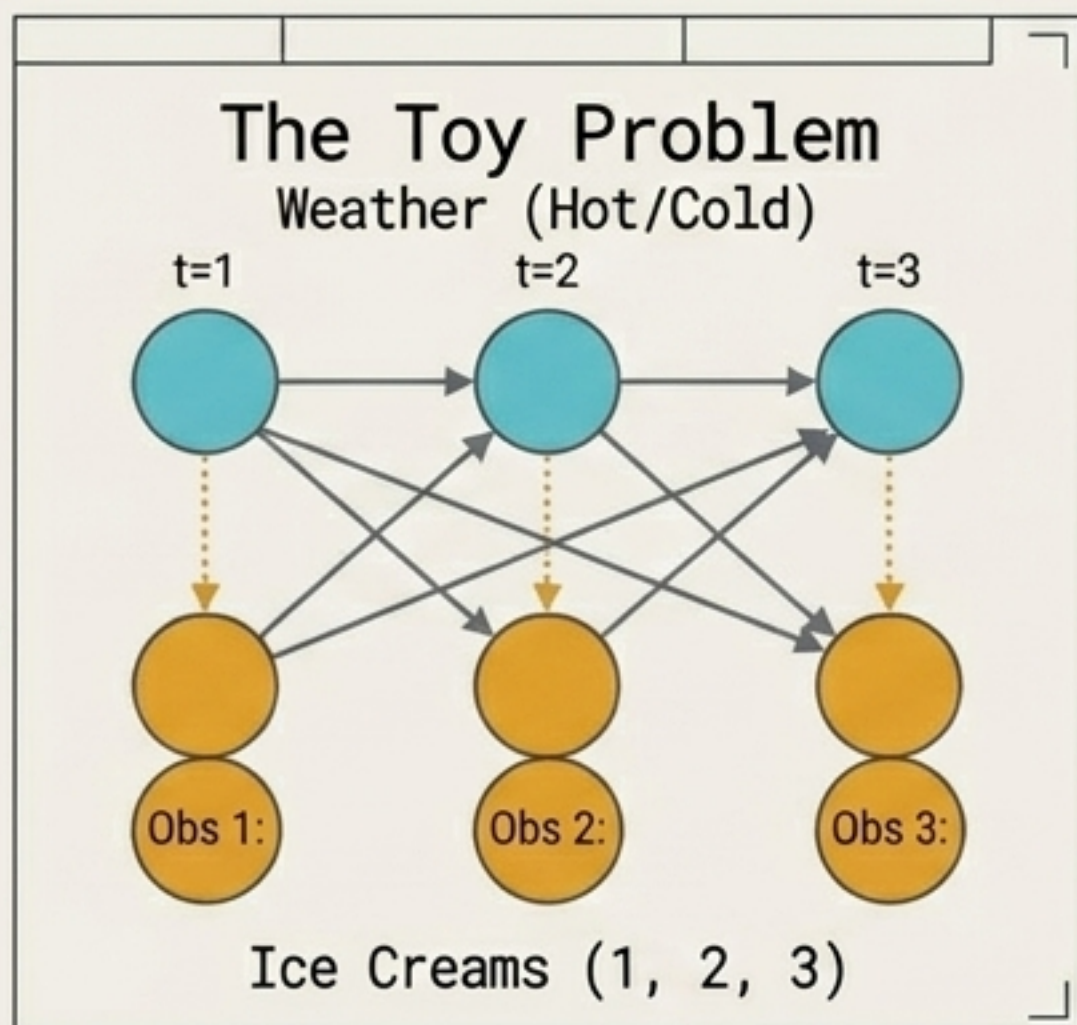
$$P(O = (3, 1, 3) | \lambda) = \alpha_3(\text{Hot}) + \alpha_3(\text{Cold})$$

$$0.023496 + 0.005066 = 0.028562$$

0.028562

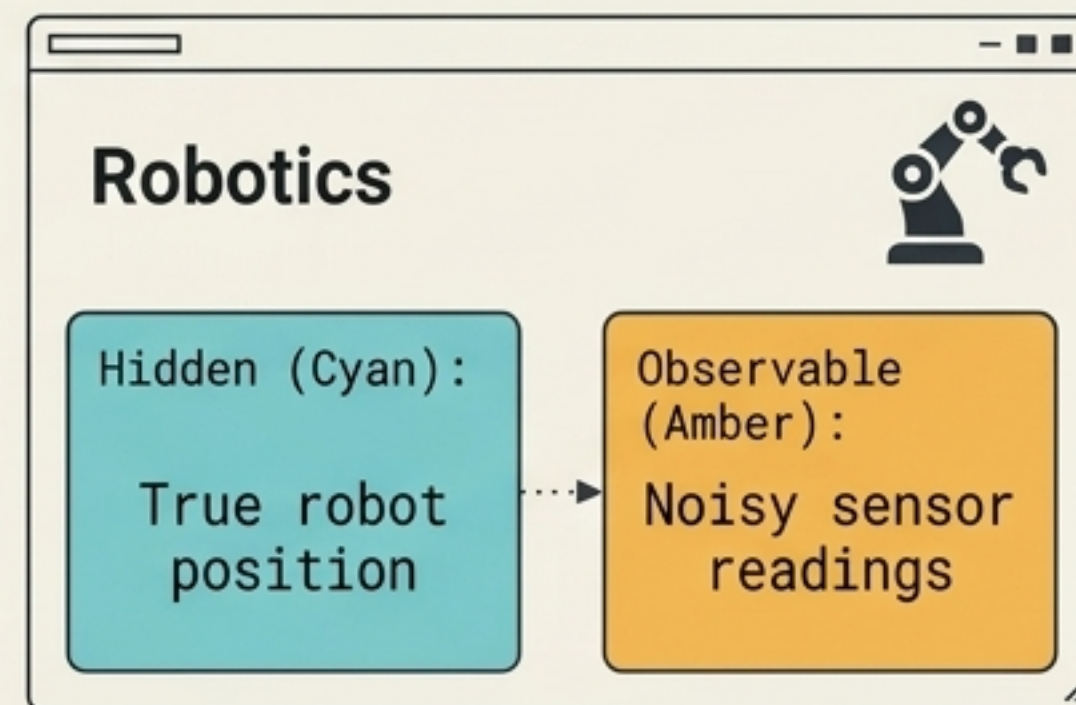
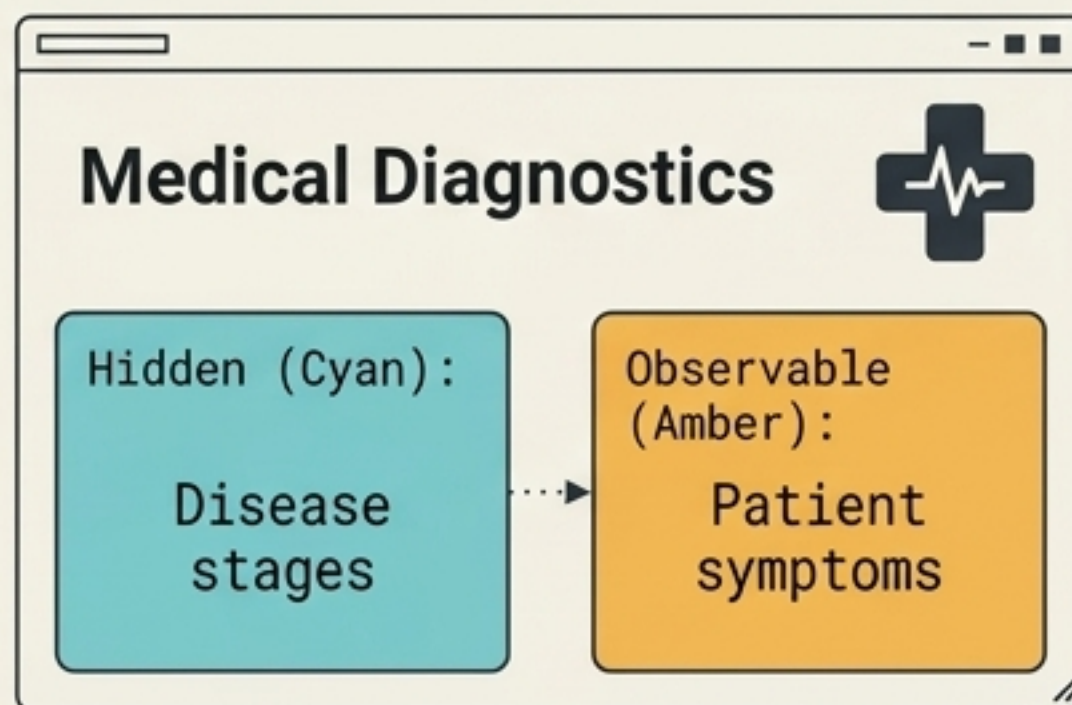
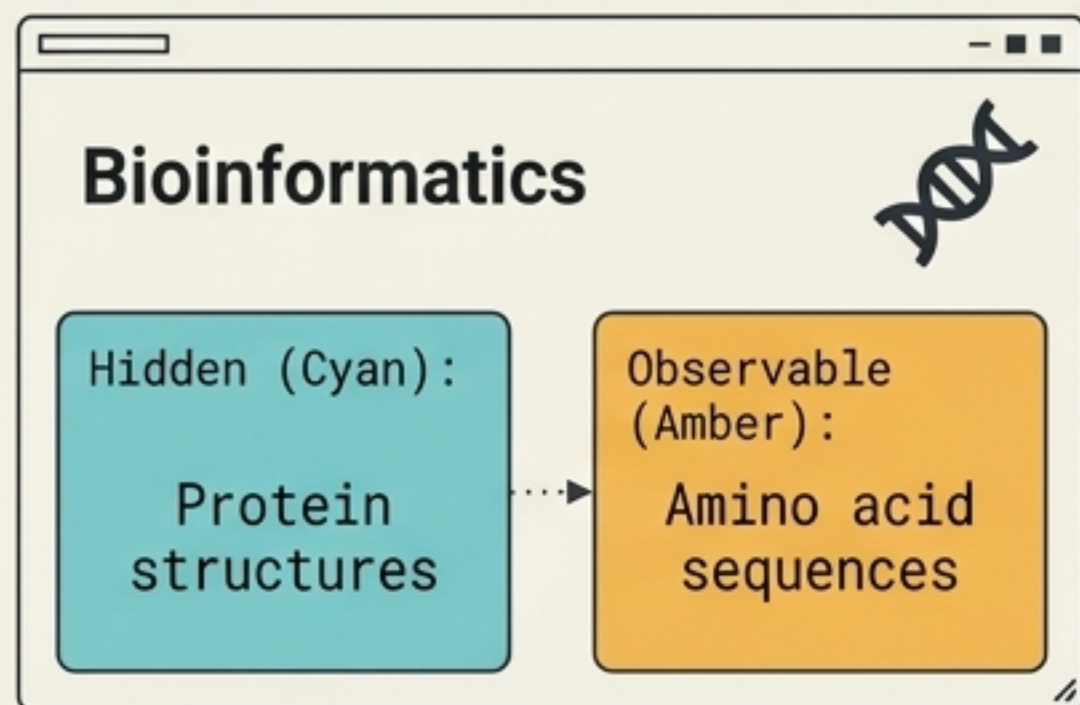
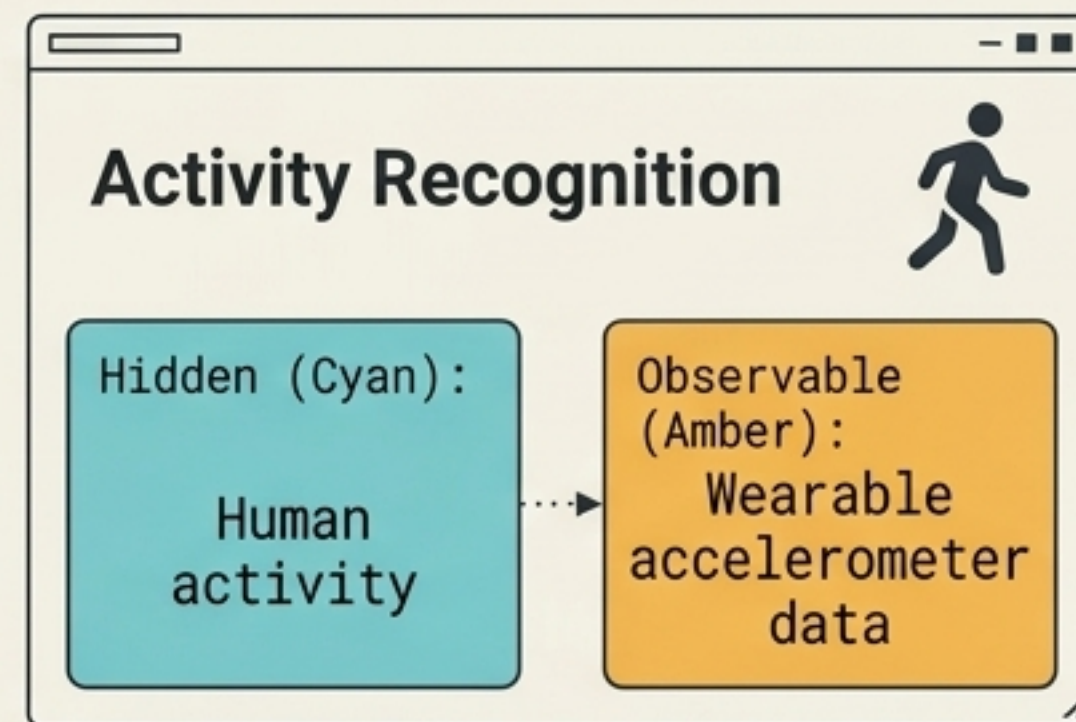
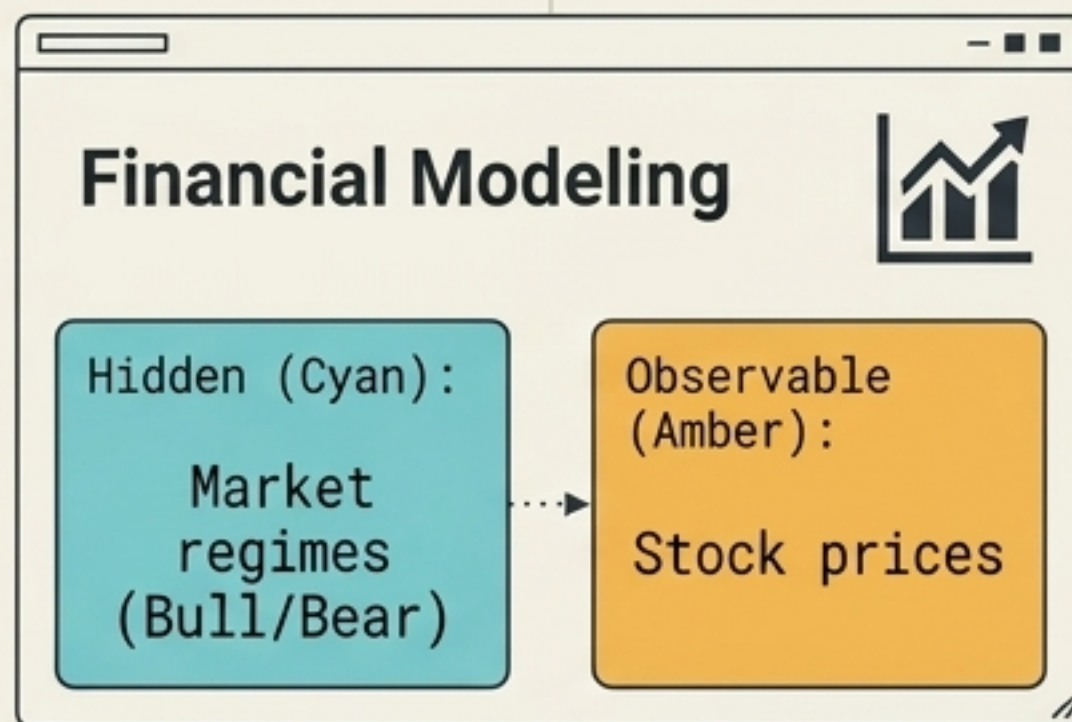
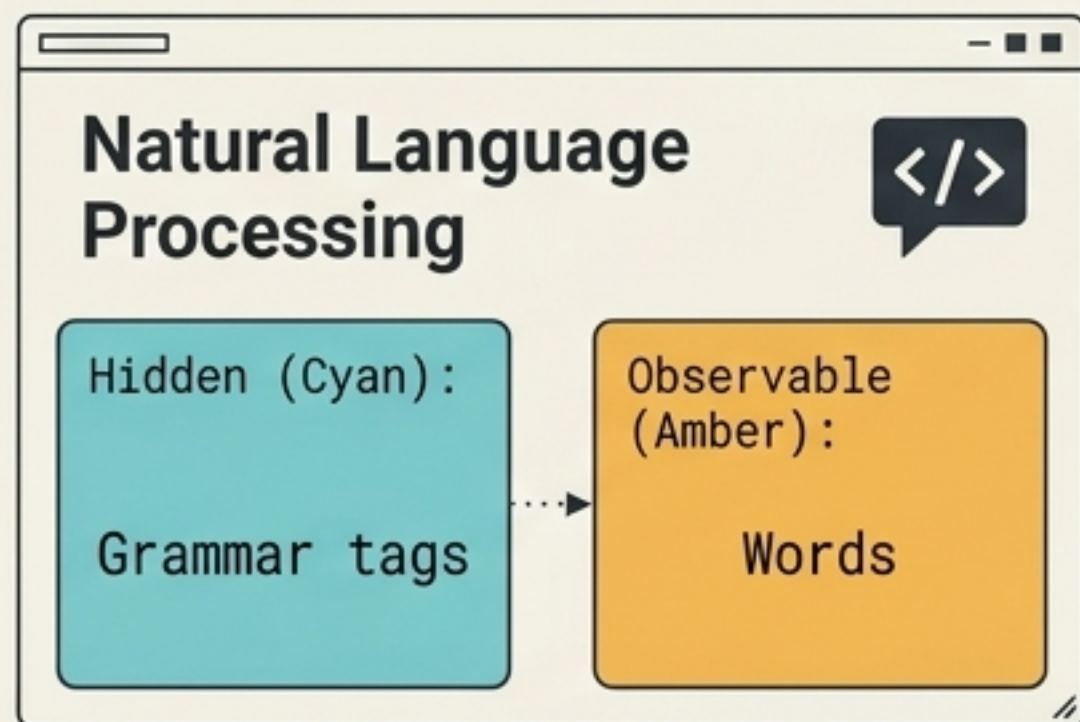
There is a 2.86% probability of observing Jason eat 3, 1, and 3 ice creams given our specific model of his habits and the weather.

The Architecture of Inference: A Universal Framework



The exact same grid. The exact same 0.028562 math. By swapping the labels, a tool for predicting ice cream becomes the engine behind Siri and DNA sequencing.

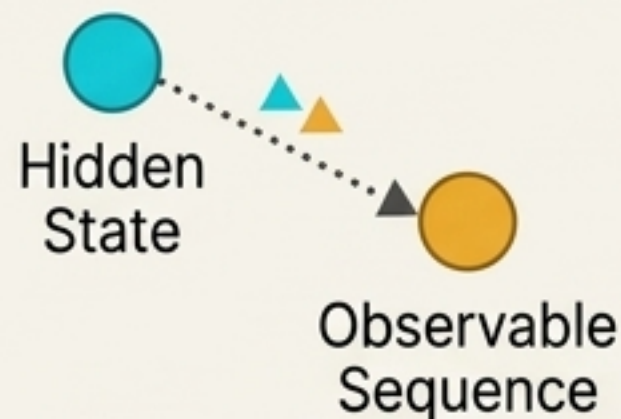
Global Deployment: Where HMMs Shine



The Indispensable Framework

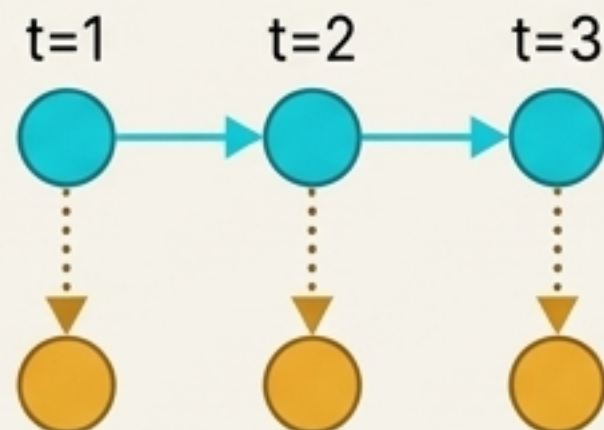
Uncovering the Unseen

Provides a rigorous mathematical framework for inferring hidden causes from observable effects where direct observation is impossible.



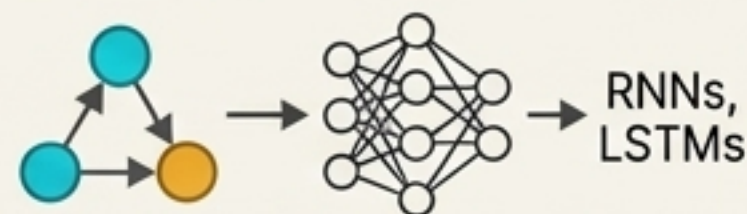
Sequential Robustness

Specifically designed to capture temporal dependencies and time-series evolution.



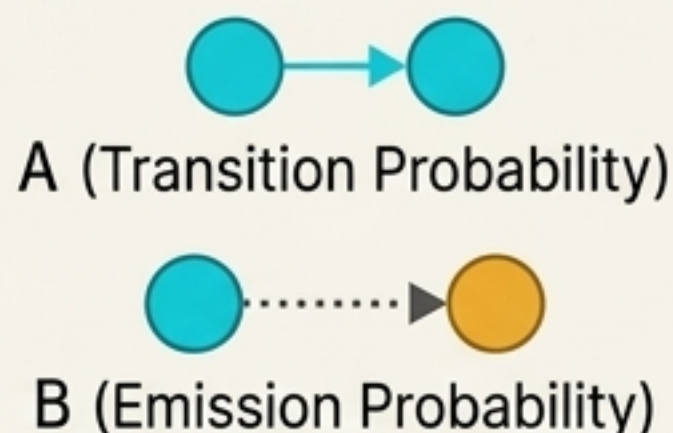
Foundation for AI

Laid the conceptual groundwork for modern deep learning sequence models like RNNs and LSTMs.



Deep Interpretability

Unlike black-box neural networks, HMM parameters (A, B) offer direct, intuitive insights into system dynamics.



**Hidden Markov Models:
Turning observable noise into actionable truth.**